

SECTION C: SQL DATABASE IMPLEMENTATION

Database Creation

The SQL script creates the database ApexCircuitDB and switches to it using the USE statement. All six tables are created with appropriate primary keys, foreign keys, and constraints.

Key Design Decisions

1. IDENTITY(1,1) is used for all primary keys to ensure auto-incrementing unique identifiers.
2. UNIQUE constraints on Email and RoleName prevent duplicate data.
3. NOT NULL constraints ensure critical fields cannot be left empty.
4. DECIMAL(10,2) for EntryFee ensures accurate currency representation.
5. The UNIQUE constraint on Results.EnrolmentID enforces the one-to-one relationship between Enrolments and Results.
6. Foreign keys use ON DELETE behavior implicitly (RESTRICT) to prevent orphaned records.

Sample Data

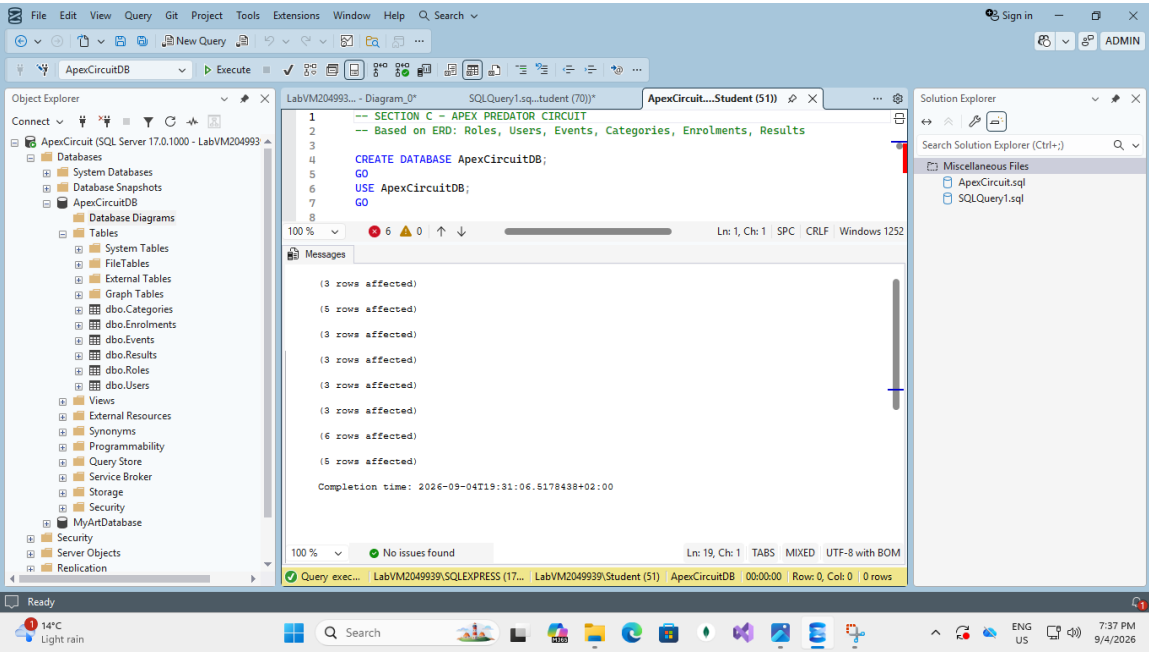
The script includes INSERT statements that seed the database with realistic South African data:

- 3 Roles: Organiser, Participant, Admin
- 5 Users: 2 Organisers (Thabo Mokoena, Sarah van Wyk), 2 Participants (Lungile Ndlovu, Johan Pretorius), 1 Admin
- 3 Events: Soweto Marathon, Cape Town Cycle Tour, Comrades Marathon
- 9 Categories: 3 per event with realistic entry fees
- 6 Enrolments: Linking participants to various categories
- 5 Results: Completed race times and positions

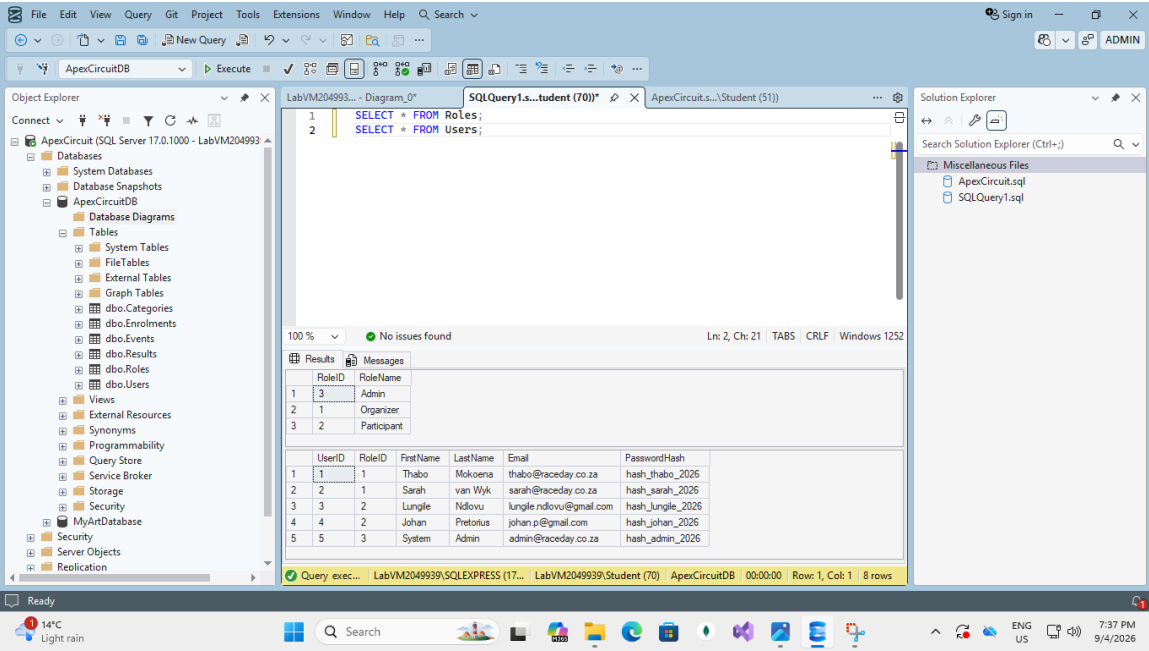
SQL Script Location

The full SQL script is available in the GitHub repository at: [docs/ApexCircuit.sql](#)

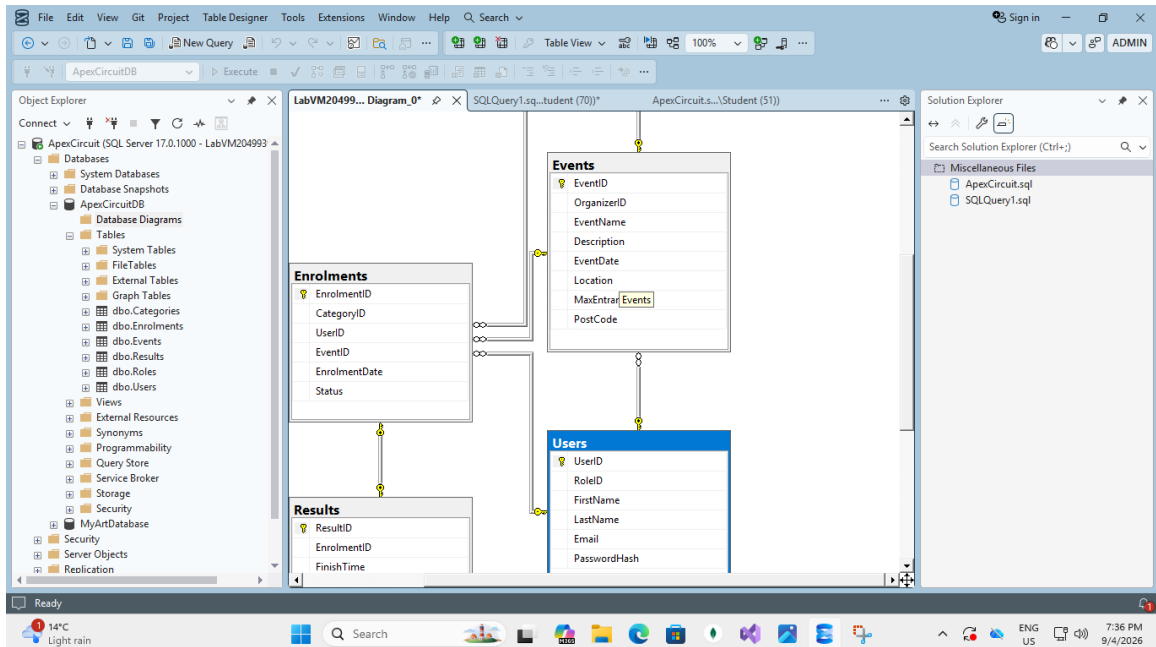
[SCREENSHOT 4: SQL Script Execution Success]

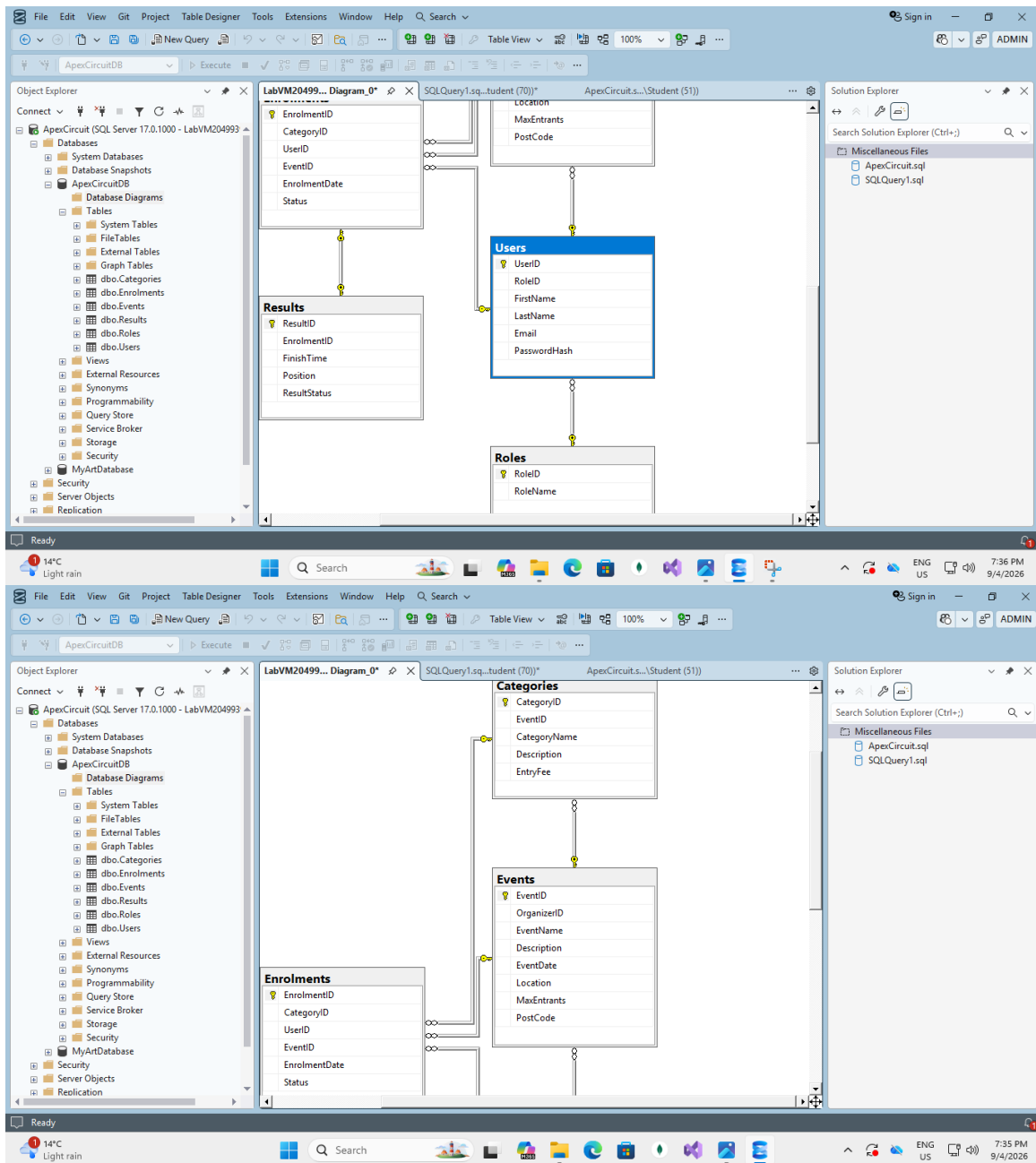


[SCREENSHOT 5: SELECT Query Results]



[SCREENSHOT 6: Database Diagram from SSMS]





LINKS & REFERENCES

GitHub Repository: <https://github.com/Tebogo4003/ApexCircuit>

YouTube Video:

Repository Contents

```
ApexCircuit/
├── .github/workflows/ci.yml    # CI/CD pipeline
├── docs/
│   ├── ERD_Apex.png          # Entity Relationship Diagram
│   ├── ApexCircuit.sql        # SQL database script
│   └── API_Endpoint_Plan.md   # API endpoint specification
├── README.md                  # Project documentation
└── Links.txt                   # Submission links
```